CH - 5 PART - V

In-Depth Study Notes: Oracle SQL Nested Summary Items and DISTINCT Operations

As an expert Oracle SQL Developer, writing efficient queries often requires multiple levels of data aggregation and strict duplicate elimination. The execution log provided (CH5-5.TXT) introduces the highly constrained architecture of **Nested Summary Items** (aggregating an aggregation) and the mechanics of the **DISTINCT** keyword.
Below is a detailed technical breakdown of these core concepts.

Concept 1: Nested Summary Items (Double Aggregation)

Core Purpose:A standard aggregate function (like AVG) groups data and returns one result per bucket. **Nested Summary Items** evaluate those grouped results and apply a second level of aggregation across them. This allows a business to answer complex questions like, "Out of all our departments' average salaries, which department's average is the absolute highest?" 1, 2.
Keyword Breakdown:

- **Aggregate Functions (MAX, MIN, SUM, AVG)**: Used in tandem, one inside the other. The inner function calculates the grouped metric, and the outer function evaluates the resulting list.
- **GROUP BY**: **Absolutely mandatory** for nested functions. It provides the initial bucketing required for the inner aggregation to generate a list of values 2.

Syntax Breakdown:

```
SELECT
outer_aggregate_function(inner_aggregate_function(target_column))
FROM table_name
GROUP BY grouping_column;
```

Comparison of Aggregation Levels:
Aggregation Type,Syntax Example,Expected Output,Oracle Requirement
Single-Level Summary,SELECT AVG(SALARY) FROM EMPLOYEES;,A single numeric value representing the whole company 1.,GROUP BY is optional.
Grouped Summary,SELECT AVG(SALARY) FROM EMPLOYEES GROUP BY DEPT_ID;,Multiple rows; one average per department 1.,GROUP BY is mandatory.
Nested Summary,SELECT MAX(AVG(SALARY)) FROM EMPLOYEES GROUP BY DEPT_ID;,A single numeric value representing the highest department average 2.,GROUP BY is mandatory 2.
Detailed Example: To determine the highest average salary paid by any single department in the HR Schema:

```
SELECT MAX(AVG(SALARY))
FROM EMPLOYEES
GROUP BY DEPARTMENT_ID;
```

Execution correctly calculates the average salary for every department internally, evaluates those averages, and outputs only the highest numeric value 1, 2.

Execution Logic:

1. **FROM**: The engine loads the EMPLOYEES table.
2. **GROUP BY**: Rows are hashed into buckets based on their DEPARTMENT_ID 2.
3. **Inner Aggregation**: The engine calculates the AVG(SALARY) for each bucket, generating an intermediate list of averages 1.
4. **Outer Aggregation**: The engine applies the MAX() function to the intermediate list, finding the highest value 2.
5. **Projection**: The single maximum value is returned.

Pro-Tips/Best Practices: Oracle enforces **three strict architectural rules** for nested summary items:

- **The GROUP BY Clause is Mandatory**: If you attempt to nest aggregates without grouping (e.g., SELECT MAX(AVG(SALARY)) FROM EMPLOYEES;), the engine will instantly throw an **ORA-00978: nested group function without GROUP BY** error 2.
- **Maximum Nesting is Two Levels**: Oracle forbids nesting three aggregates. Executing SELECT SUM(MAX(AVG(SALARY))) triggers an **ORA-00935: group function is nested too deeply** error 3.
- **Strict SELECT List Constraints**: If you use a nested aggregate, **all other items in the SELECT list must also be nested aggregates** 2. Attempting to project a single-level aggregate alongside a nested one (e.g., SELECT AVG(SALARY), MAX(AVG(SALARY))) triggers an **ORA-00979: not a GROUP BY expression** error 3.
- **The Exception**: You are legally allowed to project COUNT(*) or COUNT() on specific GROUP BY columns alongside a nested aggregate (e.g., SELECT COUNT(*), MAX(AVG(SALARY)) executes successfully) 4. However, trying to COUNT a non-grouped column like MANAGER_ID will fail with **ORA-00979** 5.

Concept 2: The DISTINCT Keyword

Core Purpose: The DISTINCT keyword acts as a data deduplicator. It forces the Oracle Engine to evaluate the projected result set and strip out any duplicate rows before displaying the data to the user 6.

Keyword Breakdown:

- **DISTINCT**: ANSI-standard keyword used to return unique values. *Why use it over the alternative?* Oracle also supports the UNIQUE keyword, which functions identically. However, DISTINCT is universally supported across all major relational databases, making your SQL code highly portable.

Syntax Breakdown:

```
-- Single column deduplication
SELECT DISTINCT column_name FROM table_name;

-- Multi-column deduplication (evaluates the combination)
SELECT DISTINCT column_1, column_2 FROM table_name;
```

Comparison of DISTINCT Behavior:

Syntax Usage,Deduplication Logic,Example Output

Single Column (DISTINCT JOB_ID),Removes duplicate identical string/number values.,Returns a list of 12 unique Job Roles that currently exist 6.
"Multi-Column (DISTINCT DEPT_ID, JOB_ID)",Removes duplicate pairs. Evaluates the combination of columns as a single tuple.,"Returns 13 unique combinations (e.g., Dept 90 + AD_PRES is treated as one unique entity) 6."
Detailed Example: To audit the HR schema and see a clean matrix of which specific Job IDs belong to which Departments, removing all employee-level redundancy:

```
SELECT DISTINCT DEPARTMENT_ID, JOB_ID
FROM EMPLOYEES;
```

Execution scans the entire employee table but collapses hundreds of rows down to just 13 unique department-to-job mappings 6.
Execution Logic:

1. **FROM**: The engine loads the table into memory 6.
2. **Projection (SELECT)**: It gathers the requested columns for every row.
3. **Sort/Hash Phase**: The engine performs an internal sort or hash aggregate on the entire result set to identify identical rows.
4. **Elimination**: Duplicate rows are discarded, and only the unique rows are passed to the user 6.

Pro-Tips/Best Practices:

- **Performance Impact**: Because DISTINCT forces the database to sort or hash the entire result set, it requires significant CPU and memory overhead. **Never use SELECT DISTINCT *** on large tables unless absolutely mathematically necessary, as it will drastically bottleneck database performance.
- **Placement Rule**: The DISTINCT keyword **must** be placed immediately after the SELECT keyword. It applies to the entire row being projected, not just the column immediately following it.